Assignment - 1

Spring 26

① $(9384)_{10}$

Binary

2	9384	
2	4692	0
2	2346	0
2	1173	0
2	586	1
2	293	0
2	146	1
2	73	0
2	36	1
2	18	0
2	9	0

BCD

9 = 1001

3 = 0011

8 = 1000

4 = 0100

$\therefore$ BCD $= (1001\ 0011\ 1000\ 0100)_{BCD}$

Ex-4

9 = 9+4 = 13 = 1101

3 = 3+4 = 7 = 0111

$$\begin{array}{r|l|l} 2 & 4 & 1 \\ 2 & 2 & 0 \\ 2 & 1 & 0 \\ & 0 & 1 \end{array}$$

$8 = 8+4 = 12 = 1100$

$4 = 4+4 = 8 = 1000$

$\therefore$ Ex-4 $= (1101\ 0111\ 1100\ 1000)_{Ex-4}$

binary $= (1001001010101000)_2$

binary $\neq$ BCD $\neq$ Ex-4 [Proved]

② $(7A3.C9)_{16} = 7\times16^2 + A\times16^1 + 3\times16^0 + C\times16^{-1} + 9\times16^{-2}$

$= (1955.785)_{10}$

$$\begin{array}{r|l|l} 5 & 1955 & \\ 5 & 391 & 0 \\ 5 & 78 & 1 \\ 5 & 15 & 3 \\ 5 & 3 & 0 \end{array}$$

$0.785 \times 5 = 3.925$ 3

$0.925 \times 5 = 4.625$ 4

$0.625 \times 5 = 3.125$ 3

0 3 1

$\therefore (1955.785)_{10} = (30310.343)_5$

③

a) $(257)_8 = 000\,010\,101\,111 = (AF)_{16}$

$(110101)_2 = 00\,110\,101 = (35)_{16}$

$$\begin{array}{r} 3A4 \\ +\ AF \\ \hline (453)_{16} \end{array} \qquad \begin{array}{r} 453 \\ -\ 35 \\ \hline (41E)_{16} \end{array}$$

$\therefore$ Ans. $(41E)_{16}$

b)

```
5433) 321654325(000040613
      31365
      -----
       50043
```

```
   45564
 ---------
   11462
    5433
 ---------
   30265
   22632
 ---------
    4333
```

$5433 \times 2 = 14166$

$5433 \times 3 = 22632$

$5433 \times 4 = 31365$

$5433 \times 5 = 40131$

$5433 \times 6 = 45564$

$5433 \times 7 = 5433$

$\therefore$ Quotient $= 40613$

Remainder $= 4333$

(4) $71 = (01000111)_2$

$-71 = -(01000111)_2 = (10111001)_{2s}$

$63 = (00111111)_2$

$-63 = -(00111111)_2 = (11000001)_{2s}$

```
    10111001
+   11000001
```

$\boxed{\cancel{1}}\ 0\ 1\ 1\ 1\ 0\ 1\ 0$

$\therefore$ overflow occurs

Now, range for 8 bits in 2s comp

$= -2^{n-1}$ to $2^{n-1}-1$

$= -2^{7}$ to $2^{7}-1$

$= -128$ to 127

$\therefore$ result exceeds the range and causes overflow

(5) banner $= (101101)_2 = (45)_{10}$

lighting $= (57)_8 = (47)_{10}$

sound system $= (2D)_{16} = (45)_{10}$

Food boxes $= (111)_2 \times 6 = 7 \times 6 = (42)_{10}$

$\therefore$ total cost $= 45 + 47 + 45 + 42 = (179)$

$\therefore$ total cost $= 45 + 44 + 45 + 42 = (176)_{10}$

$\therefore$ I have $= (1110101)_2 = (117)_{10}$

$\therefore$ money needed $= 179 - 117 = (62)_{10}$